

A Granular Ball Connectivity-Based Anomaly Factor

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Abstract—Anomaly detection technology has important application value when dealing with vast, multi-source and complex raw data, medical data is a particularly typical scenario among it, but how to identify anomalous data quickly and accurately remains a core challenge. Existing anomaly detection methods, whether based on deep learning or local density, suffer from dual limitations: high susceptibility to noise interference and substantial computational overhead. To address these issues, a granular-ball computing-based anomaly detection model is presented in this paper, named Granular Ball Connectivity-Based Anomaly Factor (GBCAF). The algorithm utilizes granular-ball to characterise data; by combining the granular-ball model with the Connectivity-based Anomaly Factor (CAF), it conducts a granular-ball density anomaly factor, in order to perform anomaly detection. In addition, this method is experimentally compared with ten other algorithms, and the results demonstrate it has good adaptability and effectiveness in medical data anomaly detection. The code is publicly available online at <https://github.com/from-china-to/GBCAF>.

Key Words—Medical data anomaly detection; multi-granularity granular-ball computing; granular-ball generation; anomaly detection

I. INTRODUCTION

Anomaly Detection (AD) focuses on finding data points with features that stand out significantly from the vast majority of the dataset. Through the rapid evolution of information technology, three key stages have marked the development of AD technology: early rule-based methods, traditional machine learning [1], and deep learning [2–4]. Although existing mainstream models have made significant progress in AD, they still face certain limitations when confronted with complex real-world scenarios. Firstly, the vast scale of data imposes stringent requirements on the computational efficiency of models; secondly, the noise interference prevalent in real-world data can easily compromise in model robustness; furthermore, the inherent ‘black box’ nature of deep learning results in a lack of transparency, making it difficult to meet the demand for model interpretability in many high-risk domains. Therefore, how to develop an AD model that balances high computational efficiency, strong anti-interference ability, and good interpretability while ensuring detection accuracy remains a pressing research priority.

To address the above challenges, the multi-granularity granular-ball (GB) [5–7] computing proposed by Wang and Xia utilizes coarse-grained ‘granular-balls’ to represent data, with a number far fewer than the number of sample points, which not only significantly improves computational efficiency [5, 8], but the model achieves exceptional noise robustness thanks to its coarse-grained nature, which minimizes susceptibility to underlying fine-grained noise [5, 6, 9]. In addition, the multi-granularity structure enables generate topological graphs and coarse-grained descriptions, which are inherently more interpretable [5, 8, 10]. Recently, research on granular-ball-based outlier detection has been further extended to areas such as mixed attribute sets, fuzzy rough sets, multi-scale density, and random walks [9–13].

The medical sector is the most typical application scenario for AD technologies that are highly efficient, robust and interpretable. With the continuous advancement of medical informatization and the continuous accumulation of clinical data, various types of medical anomaly data are difficult to be quickly and accurately identified [1, 14]. Medical institutions widely rely on the Internet of Things and various monitoring devices to collect continuous physiological signals, such as heart rate and blood pressure in real time [15], resulting in clinical data characterised by multiple sources and complexity [16]. Medical data anomalies pose a threat to patients’ health and public health safety. According to the latest patient safety report by the World Health Organization, one in every ten patients seeking medical care worldwide suffers medical harm, with over 3 million deaths annually attributable to inappropriate medical treatment [17], resulting in huge losses.

Granular-ball computing emphasizes gradually characterizing data features from coarse to fine granularities, which highly matches the cognitive approach of ‘macroscopic screening first, followed by microscopic analysis’ in clinical diagnosis. Faced with highly complex and multi-source heterogeneous medical data, granular-ball computing and the connectivity-based anomaly factor are integrated into an AD model designed for highly complex, multi-source, and heterogeneous medical datasets in this paper, named Granular Ball Connectivity-Based Anomaly Factor (GBCAF). It utilizes an unsupervised granular-

ball clustering algorithm to generate granular-balls, which significantly compresses the data scale for computation and remarkably reduces the computational time overhead of the CAF algorithm; in addition, since each granular-ball is defined solely by its center and radius, it inherently shields the CAF algorithm from high noise sensitivity; finally, the connectivity distance is calculated and a granular-ball density anomaly factor is constructed to implement medical AD. The established method extends the detection scope of traditional density detection methods and granular-ball computing.

The remainder of this paper is organized as follows. Section II proposes an AD method GBCAF based on granular-ball. Section IV introduces the experiments. Section V draws conclusions.

The main contributions of this paper are summarized as follows:

- Construct an adaptive granular-ball AD framework in the medical field, which requires no manual configuration of granularity parameters, can adapt to high-dimensional, imbalanced, and noisy clinical data, and provides an efficient and interpretable solution for medical data anomaly screening.
- A new AD model based on Granular-Ball computing and Connectivity-based Anomaly Factor (GBCAF) is proposed. This paper innovatively combines granular-ball generation with the connectivity-based anomaly factor algorithm, bringing novel and practical technical approach to risk control and AD in medical data.

II. ALGORITHM

A. GBCAF

The GBCAF first compresses the original samples into a collection of multi-granularity granular-balls, then computes the CAF anomaly score at the granular-ball level, and maps the granular-ball score back to each sample within it, as shown in Algorithm 1 and Algorithm 2.

The granular-ball generation procedure is partitioned according to the results of DM . The distribution measure DM_j is used to measure the dispersion degree of the data inside the GB_j , which follows the work of Xie et al. [7].

CAF is a connectivity-based detection algorithm, whose original idea stems from research on effective outlier detection for low-density patterns [18].

GB_j represents the j -th granular-ball, each GB_j is uniquely identified by its radius r_j and center c_j .

Definition 2.1: Let c_j refer to the center of a granular-ball, $N_k(c_j)$ be the set of k -nearest neighbor centers found in the center space, o denotes an arbitrary neighboring ball center within this neighborhood set. The connectivity-based anomaly factor ($GBCAF_k$) of GB_j is given by:

$$GBCAF_k(c_j) = \frac{|N_k(c_j)| \cdot \text{ac-dist}_{N_k(c_j)}(c_j)}{\sum_{o \in N_k(c_j)} \text{ac-dist}_{N_k(o)}(o)} \quad (1)$$

The calculation process of the ac-dist chain distance in this formula refers to the paper of Tang et al. [18], except that fine-grained sample points are replaced with coarse-grained granular-ball centers. This indicator evaluates the relative extent of connectivity between the target granular-ball and its surrounding set of granular-balls.

Definition 2.2: Let $b \in GB_j$, i.e., data point b is in the j -th GB_j , then for each data b in GB_j we have:

$$GBCAF(b) = GBCAF_k(c_j). \quad (2)$$

The GBCAF value of each point in the dataset is derived from the $GBCAF_k(c_j)$ value mapping of its located GB .

B. Time Complexity

The average time complexity of the granular-ball generation stage is $\mathcal{O}(n \cdot d \cdot \log B)$. In the anomaly detection stage, the algorithm computes the distance matrix of B granular-ball centers for neighbor search, calculates the CAF scores, and maps them to all samples, yielding a complexity of $\mathcal{O}(B^2 \cdot d + B \cdot k^2 + n)$. Accordingly, the total time complexity is $\mathcal{O}(n \cdot d \cdot \log B + B^2 \cdot d)$. Since $B \ll n$, the proposed algorithm reduces the $\mathcal{O}(n^2 \cdot d)$ overhead of the traditional CAF algorithm to $\mathcal{O}(B^2 \cdot d)$, greatly improving the efficiency on vast data.

Algorithm 1: Multi-granularity Granular-Ball Generation Process

Input : Original dataset D

Output: Multi-granularity granular-ball set GB

Initialize a root granular-ball GB_{root} with the entire dataset D ;

Utilize the distribution measure DM as the splitting criterion to recursively partition GB_{root} and its child-balls;

return the final multi-granularity granular-ball set GB ;

III. EXAMPLE DESCRIPTION

TABLE 1
Initial and standardized information system

D	a_1	a_2	a'_1	a'_2
b_0	0	0	0	0
b_1	0	2	0	0.5
b_2	0	4	0	1
b_3	10	2	1	0.5

Let $IS = (U, C)$ be the example information system listed in the left column of Table 1, where $U = (b_0, b_1, b_2, b_3)$ and $C = (a_1, a_2)$. The data obtained after Min-Max normalization preprocessing is shown in the right column of Table 1. The specific steps are as follows:

(1) First, granular-ball splitting is performed based on the normalized data listed in the table. In the initial state, all samples belong to the same root granular-ball $GB_{root} = \{b_0, b_1, b_2, b_3\}$, and its center is calculated as

Algorithm 2: GBCAF

Input : Multi-granularity granular-ball set GB

Output: The GBCAF value of each original sample b

Compute the distance matrix between all granular-ball centers in GB ; **for** each $GB_j \in GB$ **do**
 Find the k -nearest neighbor center set $N_k(c_j)$ of the center c_j ; Construct the SBN-path and calculate the average chain distance $\text{ac-dist}(c_j)$ of c_j ;
for each $GB_j \in GB$ **do**
 Calculate the mean of the ac-dist of its k -nearest neighbor centers; Calculate the granular-ball anomaly factor $GBCAF_k(c_j)$; **for** each internal sample b belonging to GB_j **do**
 Map the center score to the sample;
return the GBCAF values of all samples;

$c_{GB_{root}} = (0.2500, 0.5000)$. According to the definition, the average Euclidean distance from each point inside the ball to the center is calculated, yielding the parent ball distribution measure $DM_{GB_{root}} \approx 0.5295$. By finding the two farthest sample points b_3 and b_0 as references, GB_{root} is partitioned into two initial sub-balls: $GB_1 = \{b_3\}$ and $GB_2 = \{b_0, b_1, b_2\}$. The distribution measures of the sub-balls are calculated respectively, yielding $DM_{GB_1} = 0.0000$ (single point forming a ball) and $DM_{GB_2} \approx 0.3333$. Next, the weighted metric DM_W is calculated according to the formula:

$$DM_W = \frac{1 \times 0.0000 + 3 \times 0.3333}{4} \approx 0.2500.$$

As the splitting tolerance condition $DM_W(0.2500) < DM_{GB}(0.5295) - 10^{-6}$ is satisfied, this split takes effect. After recursive validation of the sub-balls, 3 granular-balls that meet the compactness threshold are ultimately generated, whose sample sets and final center coordinates c are as follows:

$$\begin{aligned} GB_1 &= \{b_3\}, & c_1 &= (1.0000, 0.5000), \\ GB_2 &= \{b_0, b_1\}, & c_2 &= (0.0000, 0.2500), \\ GB_3 &= \{b_2\}, & c_3 &= (0.0000, 1.0000). \end{aligned}$$

(2) Calculate the Euclidean distances between the center points of each granular-ball, and construct the center distance matrix **Dist** as follows:

$$\mathbf{Dist} = \begin{bmatrix} 0.0000 & 1.0308 & 1.1180 \\ 1.0308 & 0.0000 & 0.7500 \\ 1.1180 & 0.7500 & 0.0000 \end{bmatrix}.$$

(3) Setting the nearest-neighbour parameter $k = 2$, construct the set-based nearest path (SBN path) based on the distance matrix **Dist**. According to the weighting rule in Equation (??), when the path contains 3 nodes (2 edges

in total), the simplified decreasing weights for the two edges are $\frac{2}{3}$ and $\frac{1}{3}$ respectively. Using these weights, calculate the average chain distance (ac-dist) for each sphere center:

$$\begin{aligned} \text{ac-dist}(c_1) &= \frac{2}{3} \times \text{Dist}_{12} + \frac{1}{3} \times \text{Dist}_{23} \\ &= \frac{2}{3}(1.0308) + \frac{1}{3}(0.7500) \\ &\approx 0.9372 \end{aligned}$$

Similarly, By constructing the paths for c_2 and c_3 and calculating, we obtain:

$$\begin{aligned} \text{ac-dist}(c_2) &= \frac{2}{3}(0.7500) + \frac{1}{3}(1.0308) \approx 0.8436, \\ \text{ac-dist}(c_3) &= \frac{2}{3}(0.7500) + \frac{1}{3}(1.0308) \approx 0.8436. \end{aligned}$$

(4) Calculate the GBCAF anomaly factor for each granular-ball (i.e., the ratio of its own ac-dist to the mean ac-dist of its k neighbors). Taking GB_1 as an example:

$$\begin{aligned} \text{GBCAF}(GB_1) &= \frac{\text{ac-dist}(c_1)}{(\text{ac-dist}(c_2) + \text{ac-dist}(c_3))/2} = \frac{0.9372}{0.8436} \\ &\approx 1.1110, \end{aligned}$$

Based on the same procedure, $\text{GBCAF}(GB_2) \approx 0.9474$ and $\text{GBCAF}(GB_3) \approx 0.9474$. Since all samples within a granular-ball share the anomaly score of the ball, mapping it to the original sample space yields the final anomaly score of each object as follows:

$$\begin{aligned} \text{Score}(b_0) &\approx 0.9474, & \text{Score}(b_1) &\approx 0.9474, \\ \text{Score}(b_2) &\approx 0.9474, & \text{Score}(b_3) &= 1.1110. \end{aligned}$$

(5) Assuming a given decision threshold of $\mu = 1.0000$, among all samples, only $\text{Score}(b_3) = 1.1110 > \mu$; therefore, b_3 is an outlier in U identified by the GBCAF, whilst the scores of all other objects are below the threshold. Consequently, the final output set of outliers is $OS = \{b_3\}$.

IV. EXPERIMENTS

A. Experiment Preparation

To validate the effectiveness of the GBCAF, experiments are carried out on 11 datasets selected from the public UCI database. For the aforementioned selected datasets, this paper compares the performance of the GBCAF with 10 algorithms: Copula-Based Outlier Detection (COPOD) (2020)^[19], Weighted Neighborhood Information Network-based Outlier Detection (WNINOD) (2021)^[20], Fuzzy Information Entropy-based Outlier Detection (FIEOD) (2021)^[21], Directed density ratio Changing Rate-based Outlier Detection (DCROD) (2022)^[22], Empirical Cumulative-based Outlier Detection (ECOD) (2022)^[23], Deep Isolation Forest (DIF) (2023)^[24], Incomplete Mixed Data-Driven Outlier Detection Based on Local-Global Neighborhood Information (ILGNI) (2023)^[25], Granular-ball Fuzzy Information-based Outlier Detector (GBFG) (2025)^[26], Fuzzy Granule Density-based Anomaly

Detection (GDAD) (2025)^[27], and Fuzzy Granule Density-Based Outlier Detection With Multi-Scale granular-balls (MGBOD_tkde) (2025)^[10].

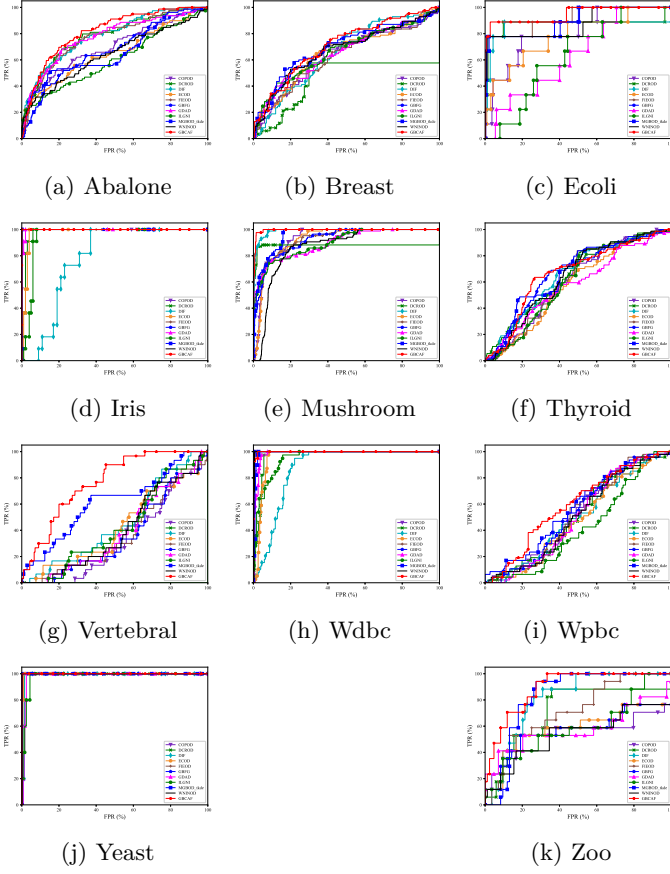


Fig. 1. Experimental comparison results on ROC curves.

We conduct experiments on 11 datasets from the public UCI database¹², which are collected from medical and biomedical fields, including Abalone_variant1 (Abalone), Breast_cancer_variant1 (Breast), Ecoli (Ecoli), Iris_irisvirginica_11_variant1 (Iris), Mushroom_p_85_variant1 (Mushroom), Thyroid_disease_variant1 (Thyroid), Vertebral (Vertebral), Wdbc_m_39_variant1 (Wdbc), Wpbc_variant1 (Wpbc), Yeast_eri_5_variant1 (Yeast), and Zoo_variant1 (Zoo).

B. Analysis of Experimental Results

The ROC curves of the 11 detection algorithms are displayed in Figure 1, among which the red five-pointed star curve represents the GBCAF proposed. The ROC curves of 10 comparison algorithms show a monotonically rising trend, as shown in Figure 1. The ROC curve can provide an intuitive comparison of different detection algorithms. Table 2 shows the comparison results of AUC values of various AD algorithms on each attribute dataset, where

the highest scores are displayed in boldface. From the table data, it can be intuitively seen that: the global average AUC of the proposed GBCAF model is as high as 0.8534, and its overall performance on all 11 datasets is significantly better than the other 10 comparison algorithms; especially on medical-specific datasets such as breast tumors and thyroid diseases, its advantage is more prominent. The experimental outcomes fully verify the feasibility of the scheme integrating granular-ball computing with the connectivity anomaly factor, and this framework is adaptable to high-dimensional, imbalanced, and noisy clinical medical data. Table 3 shows the experimental outcomes of F1-score, with the best outcomes indicated in boldface. From Table 3, it can be seen that the GBCAF proposed achieved the best results on 8 datasets. These results indicate that GBCAF has advantages in both overall performance and statistical significance.

V. CONCLUSION

AD in medical data is a core research challenge in the field of smart medical. To address the shortcomings of traditional deep learning and density-based algorithms in clinical applications, such as weak noise resistance, poor interpretability, and high computational cost, this paper proposes the GBCAF AD model for medical scenarios, combining the adaptive granular-ball generation mechanism with the Connectivity-based Anomaly Factor (CAF). Sufficient comparative experiments were conducted against 10 mainstream AD algorithms on 11 sets of UCI standard datasets covering clinical applications such as oncology and thyroid. The results demonstrate that the GBCAF model can more accurately capture subtle clinical anomaly features, effectively suppress false detection problems caused by equipment noise, and significantly reduce the computational overhead of traditional large-scale data. This study improves operational efficiency and stability while ensuring detection accuracy, providing an efficient and reliable new solution for the practical application of granular-ball computing theory in intelligent auxiliary diagnosis and medical data quality control.

¹<https://github.com/from-china-to/Outlier-detection>

²<https://github.com/Belloney/Outlier-detection>

TABLE 2
EXPERIMENTAL EVALUATION OF AUC

Datasets	GBCAF	COPOD	DCROD	DIF	ECOD	FIEOD	GBFG	GDAD	ILGNI	MGBOD_tkde	WNINOD
Abalone	0.8149	0.6958	0.7803	0.7874	0.6531	0.7702	0.6806	0.7724	0.6040	0.6030	0.6410
Breast	0.7090	0.6322	0.6035	0.6539	0.6555	0.6321	0.6713	0.6536	0.6587	0.6644	0.6707
Ecoli	0.9480	0.8087	0.8746	0.8637	0.7808	0.9168	0.8991	0.6351	0.6381	0.8940	0.8739
Iris	1.0000	1.0000	0.9827	0.7791	0.9773	1.0000	1.0000	0.9973	0.9509	1.0000	1.0000
Mushroom	0.9954	0.9283	0.9251	0.9887	0.9220	0.9323	0.9218	0.8988	0.8979	0.9212	0.8664
Thyroid	0.6661	0.6110	0.6461	0.6590	0.5808	0.6120	0.6639	0.5840	0.6096	0.6385	0.6396
Vertebral	0.7684	0.3349	0.5000	0.4590	0.4200	0.3690	0.3595	0.3884	0.4244	0.5613	0.3956
Wdbc	0.9986	0.9956	0.9682	0.8705	0.9591	0.9986	0.9976	0.9849	0.9512	0.9836	0.9964
Wdbc	0.5962	0.5226	0.5129	0.5013	0.4810	0.5247	0.5305	0.5071	0.4210	0.5211	0.5064
Yeast	0.9998	0.9936	0.9898	0.9870	0.9952	0.9912	0.9996	0.9842	0.9810	0.9912	0.9982
Zoo	0.8915	0.5497	0.7521	0.8109	0.5819	0.7143	0.5630	0.5798	0.5890	0.3417	0.5385
Average 1st order	0.8534 11	0.7339 1	0.7759 0	0.7600 0	0.7279 0	0.7692 2	0.7534 1	0.7260 0	0.7023 0	0.7382 1	0.7388 1

TABLE 3
EXPERIMENTAL EVALUATION OF F1-SCORE

Datasets	GBCAF	COPOD	DCROD	DIF	ECOD	FIEOD	GBFG	GDAD	ILGNI	MGBOD_tkde	WNINOD
Abalone	0.1763	0.1805	0.1558	0.2353	0.2000	0.1572	0.1789	0.1895	0.2000	0.1132	0.1905
Breast	0.5560	0.5205	0.5052	0.5349	0.5381	0.4957	0.5439	0.5000	0.5198	0.5556	0.5430
Ecoli	0.8000	0.2963	0.6667	0.5000	0.2963	0.7368	0.7368	0.1290	0.0952	0.7059	0.7368
Iris	1.0000	1.0000	0.8333	0.3810	0.8462	1.0000	1.0000	0.9167	0.7407	1.0000	1.0000
Mushroom	0.8435	0.3125	0.6792	0.7222	0.3107	0.5345	0.4103	0.5821	0.5254	0.4513	0.2000
Thyroid	0.0438	0.0236	0.0485	0.0330	0.0225	0.0267	0.0310	0.0253	0.0258	0.0410	0.0266
Vertebral	0.4040	0.3014	0.2247	0.2615	0.2857	0.3066	0.3077	0.2841	0.2707	0.3125	0.2933
Wdbc	0.9600	0.9512	0.7835	0.4774	0.7379	0.9744	0.9750	0.8315	0.6667	0.8941	0.9750
Wdbc	0.4528	0.4162	0.4205	0.4053	0.4188	0.4327	0.4231	0.4242	0.4410	0.4233	0.4167
Yeast	0.9091	0.7692	0.4000	0.3704	0.5333	0.8000	0.9091	0.2667	0.2667	0.8333	0.8333
Zoo	0.6667	0.4500	0.4839	0.5306	0.4500	0.4865	0.4286	0.4667	0.3871	0.5778	0.3810
Average 1st order	0.6193 8	0.4747 1	0.4728 1	0.4047 1	0.4218 0	0.5410 1	0.5404 3	0.4196 0	0.3763 0	0.5371 1	0.5087 2

REFERENCES

- [1] L. Mei, F. Zhang, and Q. Gao, "Survey of outlier detection techniques," *Application Research of Computers*, vol. 37, no. 12, pp. 3521–3527, 2020.
- [2] L. Ruff, J. R. Kauffmann, R. A. Vandermeulen, G. Montavon, W. Samek, M. Kloft, T. G. Dietterich, and K.-R. Müller, "A unifying review of deep and shallow anomaly detection," *Proceedings of the IEEE*, vol. 109, no. 5, pp. 756–795, 2021.
- [3] G. Pang, C. Shen, L. Cao, and A. van den Hengel, "Deep learning for anomaly detection," *ACM Computing Surveys*, vol. 54, no. 2, pp. 1–38, 2021.
- [4] Z. Wu, S. Pan, F. Chen, G. Long, C. Zhang, and P. S. Yu, "A comprehensive survey on graph neural networks," *IEEE Transactions on Neural Networks and Learning Systems*, vol. 32, no. 1, pp. 1–24, 2021.
- [5] S. Xia, G. Wang, X. Gao, X. Lian, and H. Kuai, "Granular-ball computing: an efficient, robust, and interpretable adaptive multi-granularity representation and computation method," arXiv:2304.11171, 2023.
- [6] S. Xia, Y. Liu, X. Ding, G. Wang, H. Yu, and Y. Luo, "Granular ball computing classifiers for efficient, scalable and robust learning," *Information Sciences*, vol. 483, pp. 136–152, 2019.
- [7] J. Xie, W. Kong, S. Xia, G. Wang, and X. Gao, "An efficient spectral clustering algorithm based on granular-ball," *IEEE Transactions on Knowledge and Data Engineering*, vol. 35, no. 9, pp. 9743–9753, 2023.
- [8] G. Wang, "Dgcc: Data-driven granular cognitive computing," *Granular Computing*, vol. 2, no. 4, pp. 343–355, 2017.
- [9] X. Su, Z. Yuan, B. Chen, D. Peng, H. Chen, and Y. Chen, "Detecting anomalies with granular-ball fuzzy rough sets," *Information Sciences*, vol. 678, p. 121016, 2024.
- [10] C. Gao, X. Tan, J. Zhou, W. Ding, and W. Pedrycz, "Fuzzy granule density-based outlier detection with multi-scale granular balls," *IEEE Transactions on Knowledge and Data Engineering*, vol. 37, no. 3, pp. 1182–1197, 2025.
- [11] "Fuzzy granular anomaly detection using markov random walk," *Information Sciences*, vol. 646, p. 119400, 2023.
- [12] C. Liu, Z. Yuan, Y. Yuan, H. Chen, D. Peng, and X. Song, "Fuzzy knn entropy and its anomaly detection," *IEEE Transactions on Knowledge and Data Engineering*, pp. 1–14, 2025.
- [13] S. Wang, Z. Yuan, S. Cheng, H. Chen, and D. Peng, "Granular-ball computing-based random walk for anomaly detection," *Pattern Recognition*, vol. 165, p. 111588, 2025.
- [14] V. Chandola, A. Banerjee, and V. Kumar, "Anomaly detection," *ACM Computing Surveys*, vol. 41, no. 3, pp. 1–58, 2009.
- [15] M. M. Khan and M. Alkhathami, "Anomaly detection in iot-based healthcare: machine learning for enhanced security," *Scientific Reports*, vol. 14, p. 5872, 2024.
- [16] B. Shickel, P. J. Tighe, A. Bihorac, and P. Rashidi, "Deep ehr: A survey of recent advances in deep learning techniques for electronic health record (ehr) analysis," *IEEE Journal of Biomedical and Health Informatics*, vol. 22, no. 5, pp. 1589–1604, 2018.
- [17] World Health Organization, "Global patient safety action plan 2021-2030: Towards eliminating avoidable harm in health care," Geneva, 2021. [Online]. Available: <https://www.who.int/publications/i/item/9789240032705>
- [18] J. Tang, Z. Chen, A. W.-C. Fu, and D. W. Cheung, "Enhancing effectiveness of outlier detections for low density patterns," in *Advances in Knowledge Discovery and Data Mining: 6th Pacific-Asia Conference, PAKDD 2002*. Springer, 2002, pp. 535–548.
- [19] Z. Li, Y. Zhao, N. Botta, C. Ionescu, and X. Hu, "Copod: Copula-based outlier detection," in *2020 IEEE International Conference on Data Mining (ICDM)*. IEEE, 2020, pp. 1118–1123.
- [20] Y. Wang and Y. Li, "Outlier detection based on weighted neighbourhood information network for mixed-valued datasets," *Information Sciences*, vol. 564, pp. 396–415, 2021.
- [21] Z. Yuan, H. Chen, T. Li, J. Liu, and S. Wang, "Fuzzy information entropy-based adaptive approach for hybrid feature outlier detection," *Fuzzy Sets and Systems*, vol. 421, pp. 1–28, 2021.
- [22] K. Li, X. Gao, S. Fu, X. Diaoy, P. Ye, B. Xue, J. Yu, and Z. Huang, "Robust outlier detection based on the changing rate of directed density ratio," *Expert Systems with Applications*, vol. 207, p. 117988, 2022.
- [23] Z. Li, Y. Zhao, X. Hu, N. Botta, C. Ionescu, and G. H. Chen, "Ecod: Unsupervised outlier detection using empirical cumulative distribution functions," *IEEE Transactions on Knowledge and Data Engineering*, vol. 35, no. 12, pp. 12 181–12 193, 2023.
- [24] H. Xu, G. Pang, Y. Wang, and Y. Wang, "Deep isolation forest for anomaly detection," *IEEE Transactions on Knowledge and Data Engineering*, vol. 35, no. 12, pp. 12 591–12 604, 2023.
- [25] R. Li, H. Chen, S. Liu, X. Li, Y. Li, and B. Wang, "Incomplete mixed data-driven outlier detection based on local-global neighbourhood information," *Information Sciences*, vol. 633, pp. 204–225, 2023.
- [26] Q. Li, Z. Yuan, D. Peng, X. Song, H. Zheng, and X. Su, "Granular-ball fuzzy information-based outlier detector," *International Journal of Approximate Reasoning*, vol. 185, p. 109473, 2025.
- [27] B. Chen, Z. Yuan, D. Peng, X. Chen, H. Chen, and Y. Chen, "Integrating granular computing with density estimation for anomaly detection in high-dimensional heterogeneous data," *Information Sciences*, vol. 690, p. 121566, 2025.